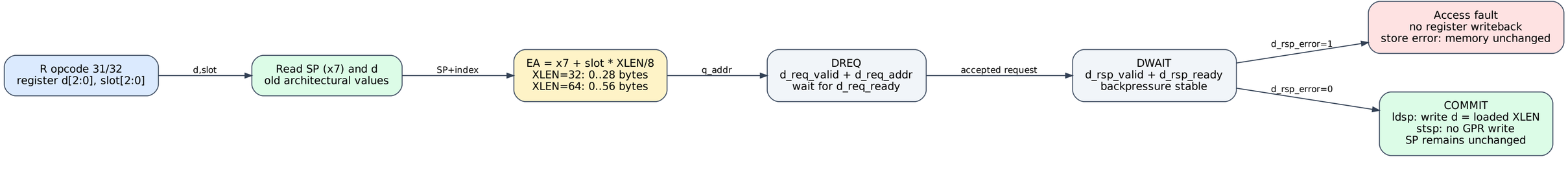
MICA 2.0 | stack loads/stores and precise retirement
R opcode 31/32
register d[2:0], slot[2:0]
d,slot
Read SP (x7) and d
old architectural values
SP+index
EA = x7 + slot * XLEN/8
XLEN=32: 0..28 bytes
XLEN=64: 0..56 bytes
q_addr
DREQ
d_req_valid + d_req_addr
wait for d_req_ready
accepted request
DWAIT
d_rsp_valid + d_rsp_ready
backpressure stable
d_rsp_error=1
Access fault
no register writeback
store error: memory unchanged
d_rsp_error=0
COMMIT
ldsp: write d = loaded XLEN
stsp: no GPR write
SP remains unchanged